SA 1: Annotation Workflow Report

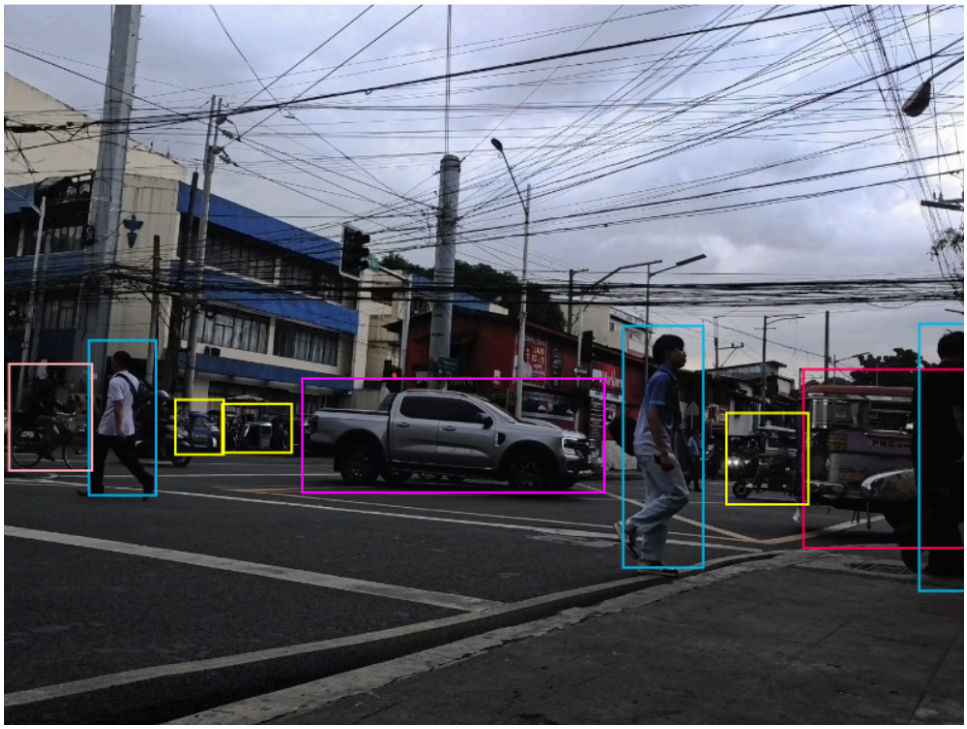
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I. Project Context and Gap

This project operates within the transportation domain and is oriented toward constructing a video-derived computer vision dataset that accurately reflects the vehicle types and traffic conditions characteristic of urban Philippine road environments. A persistent limitation in the machine learning literature on road-scene detection is the geographic and contextual bias of widely used benchmark datasets, most of which have been collected in North American, European, or East Asian settings. These datasets tend to encode vehicle fleets, lane configurations, and road geometries that diverge substantially from those observed in Philippine urban traffic (Everingham et al., 2010; Vasquez Hidalgo et al., 2024), resulting in models that may exhibit degraded generalization when applied to local deployment contexts (Bochkovskiy et al., 2024), particularly when the relevant vehicle categories are absent or insufficiently represented in the training labels.

The gap that motivates this dataset is one of representation and class separability. Existing datasets frequently collapse distinct local vehicle types into coarse superclasses (e.g., a single "vehicle" label), which obscures class-specific visual patterns and limits the diagnostic utility of trained detectors. This dataset is designed to address that gap by capturing fine-grained, locally relevant vehicle categories under conditions of visual diversity: varying illumination, partial occlusion, and mixed traffic density, so that a YOLO-style object detector can learn discriminative representations rather than relying on overgeneralized labels. The broader contribution of this work is a structured, reproducible dataset pipeline, extending from raw video acquisition to versioned, annotated frames, that supports downstream modeling while adhering to ethical data practices and maintaining methodological transparency.

II. Dataset Description

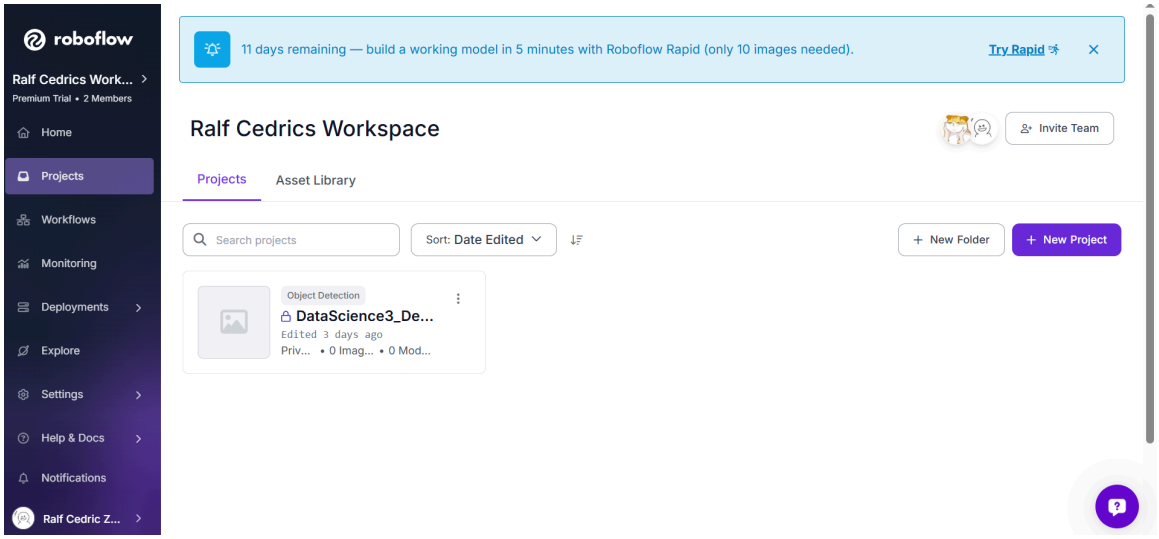


The dataset was derived from self-gathered roadside video footage recorded in an urban Philippine setting. Footage was captured along Kalayaan Avenue, Makati, Metro Manila, at approximately 3:00 PM on a weekday afternoon—a time period characterized by active mixed traffic involving both private vehicles and locally prevalent public utility or niche transport types. The original recording configuration has been documented to ensure reproducibility: the raw footage spans an initial duration of 45 minutes at a native frame rate of 60 frames per second (fps) and a resolution of 1080p. For dataset construction purposes, the footage was trimmed to a final duration of 30 minutes, retaining the segments most representative of the target vehicle categories while eliminating redundant intervals that would otherwise inflate the proportion of near-duplicate frames.

The methodological decision to use a single continuous clip, rather than aggregating multiple shorter recordings, was made to preserve temporal continuity and facilitate systematic sampling during the frame extraction stage. It should be noted that the resulting dataset is conceptualized as an image-based detection dataset—composed of individually annotated still frames—rather than a temporal tracking dataset. The primary deliverable is therefore a structured directory of frames paired with corresponding annotation files in a YOLO-compatible format. Environmental

variability factors such as weather conditions, dynamic shadow patterns, camera stability, and fluctuations in traffic density are acknowledged as features of the captured scene but are not explicitly controlled or stratified in this release.

III. Tool Justification



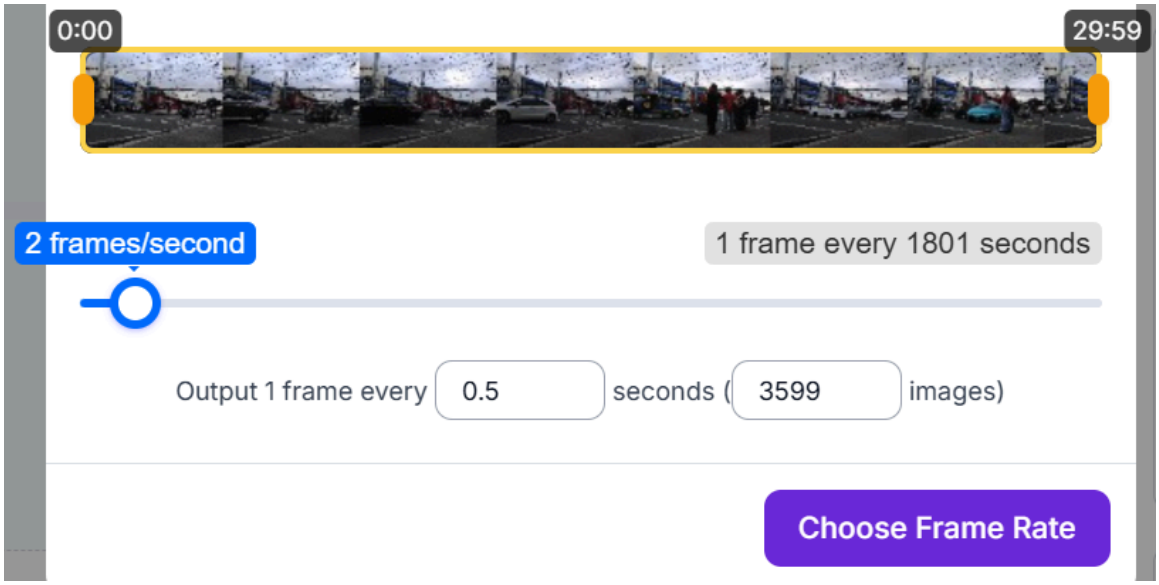
Roboflow was selected as the primary annotation and dataset management platform on the basis of its end-to-end support for video-derived dataset workflows (Roboflow, 2024), which directly aligns with the deliverable requirements of this activity. Specifically, the platform facilitates: (1) video ingestion and automated frame extraction at configurable sampling rates; (2) browser-based bounding box annotation with multi-class label support; (3) dataset-level analytics for quality inspection and class distribution review; (4) version-controlled dataset snapshots that record the exact preprocessing, augmentation, and split configuration applied at each release; and (5) export in formats natively compatible with YOLO training pipelines.

The selection of Roboflow is further justified on grounds of reproducibility and auditability. Dataset versioning allows the project team to record and reconstruct the precise state of the dataset at any given point in the development process, reducing the risk of undocumented modifications that could compromise the integrity of model evaluation. Additionally, Roboflow's incremental development workflow is particularly well-suited to a student-built dataset: annotation can begin with an initial labeled subset, label definitions can be refined iteratively, and the dataset can be scaled up while preserving a traceable history of all configuration changes across versions. This approach is consistent with the transparent, stepwise documentation expected of

dataset-building deliverables, in which the process of construction is as important as the final annotated output.

IV. The Annotation Workflow

Uploading the Video & Sampling Rate



The video upload stage fulfills two primary functions: ingesting the raw footage into a managed annotation environment and defining the temporal sampling rate that governs the density of frame extraction. In this project, the trimmed 30-minute clip was uploaded to Roboflow and sampled at a rate of 2 frames per second, yielding a total of 3,599 candidate images. The selection of a 2 fps sampling rate was determined through frame-by-frame inspection of a preliminary extraction, during which the project team evaluated whether each target class could be reliably observed with sufficient visual clarity and whether the resulting frame set retained adequate diversity without introducing excessive near-duplicate redundancy. A higher sampling rate was considered but deemed unnecessary given the relatively stable camera position and the moderate pace of vehicle movement within the field of view.

The Class Labels

The dataset adopts a label taxonomy of seven classes, derived from the vehicle categories observed within the recorded scene. The class structure is designed to

reflect the composition of Philippine urban traffic and to avoid collapsing visually and functionally distinct vehicle types into a generic superclass. Such label collapsing is methodologically problematic because it conceals class-specific error modes—for instance, a model that misclassifies jeepneys as trucks would not be identified as erroneous under a single "vehicle" label, and reduces the practical value of the dataset for deployment in Philippine road contexts. The seven labeled classes, along with their operational definitions, are presented in the table below.

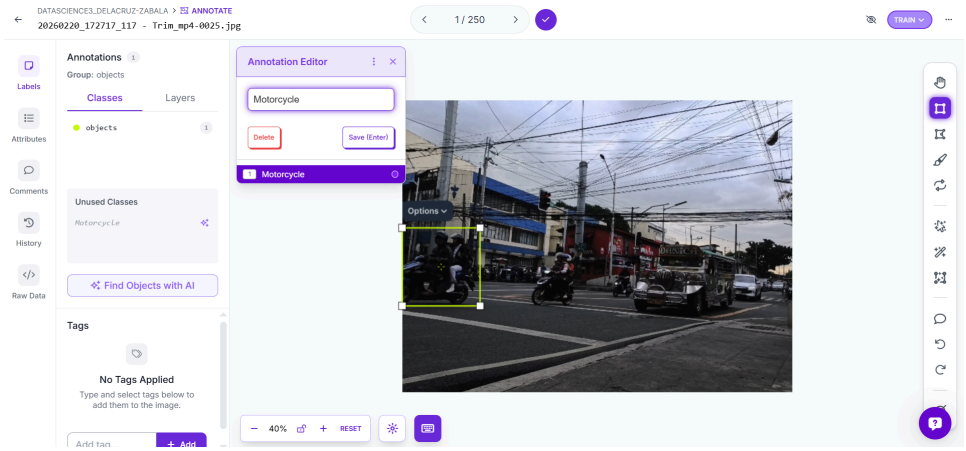
Class Label	Description
Bicycle	A man-powered bicycle or a motorcycle with a visible frame and handlebars
Car	A standard passenger sedan/hatchback/compact vehicle with a smaller height than SUVs.
Jeepney	A Philippine public utility vehicle with elongated body, open/partially open side or rear entry, and distinctive “jeep-like” front profile, commonly used for public transport.
Pedestrian	A person walking, standing, crossing, or waiting on/near the roadway, including those at sidewalks, crossings, or roadside edges.
SUV	A taller passenger vehicle than a car, typically with a larger cabin, higher ride height, bulkier body, and often a more vertical rear profile.
Tricycle	A Philippine motorized tricycle: a motorcycle with an attached sidecar/cabin, forming a three-wheeled public transport vehicle with a distinct sidecar outline.
Truck	A commercial or utility vehicle designed for cargo transport, typically larger than cars/SUVs, with a cargo bed, container body, or distinct commercial rear structure.

Frame Annotation

Annotation was performed using rectangular bounding boxes, consistent with the input format required by YOLO-style object detectors. Bounding box annotation is appropriate for this task because YOLO architectures learn localization and classification jointly from rectangular region proposals, and the selected label classes

are all physically bounded, spatially coherent objects amenable to tight rectangular enclosure (Redmon et al., 2016).

1. **Bounding Box Consistency Rules**



To ensure annotation consistency across the dataset, annotators adhered to a shared bounding box protocol developed prior to the main labeling phase. Boxes were drawn to enclose the visible extent of each target object as tightly as possible, minimizing the inclusion of background pixels while fully covering the object's silhouette when unobstructed. For pedestrian instances, boxes were placed to encompass the visible body region; for vehicles, boxes enclosed the vehicle body as observed within the frame. In cases where objects were partially truncated by the image boundary, boxes were extended only to the frame edge, without any attempt to estimate or extrapolate the unseen portion of the object.

2. **Handling Difficult and Low-Quality Cases**

Difficult annotation cases were resolved conservatively, in accordance with the principle of minimizing label noise over maximizing instance count. Frames in which substantial occlusion prevented reliable object identification were removed from the dataset during the post-annotation review process. Similarly, vehicles that appeared at sufficient distance to preclude confident class assignment were excluded from labeling, as imposing labels under conditions of inadequate visual resolution constitutes a source of systematic ground-truth error. To further maintain annotation reliability, labeling was restricted to objects within the visually informative region of the scene—specifically, those observable up to the pedestrian crossing and intersection area, where vehicle morphology

and class-discriminative visual cues remained sufficiently legible for consistent classification.

3. Inter-Annotator Agreement and Professional Oversight

INTER-ANNOTATOR AGREEMENT AND LABEL VALIDATION LETTER

February 28, 2026

To Whom It May Concern,

We are Computer Science students of **Mapúa University** under the course **CSS182-3/BM2 - Data Science 3**, hereby certify that we conducted a structured annotation and validation process for our video-derived dataset in compliance with the course requirements.

Our selected domain is **Transportation**, specifically **vehicle classification**, and the dataset annotations were created for the following seven (7) labels:

- bicycle
- car
- suv
- jeepney
- truck
- tricycle
- pedestrian

As part of our **quality assurance protocol**, we implemented an **inter-annotator review process** to verify the correctness and consistency of labels applied to extracted video frames. The annotation outputs were reviewed by at least two (2) annotators and checked for agreement in terms of:

- **Class label correctness** (proper object category assigned)
- **Consistency of label usage** across similar objects and scenes
- **Handling of ambiguous cases** (such as occlusion, motion blur, low-light frames, partial object visibility)
- **Annotation quality standards** based on our agreed labeling guidelines

After review and reconciliation of disagreements, the group confirms that the final labels included in the submitted structured annotated dataset have been **checked, validated, and approved as correct to the best of our knowledge** under the defined annotation guidelines of this project.

This letter serves as documentation that an inter-annotator validation step was completed as part of the dataset preprocessing and quality assurance stage.

We affirm that this process was performed in good faith and that all corrections identified during the review were incorporated into the final dataset version submitted for course requirements.

Respectfully submitted,

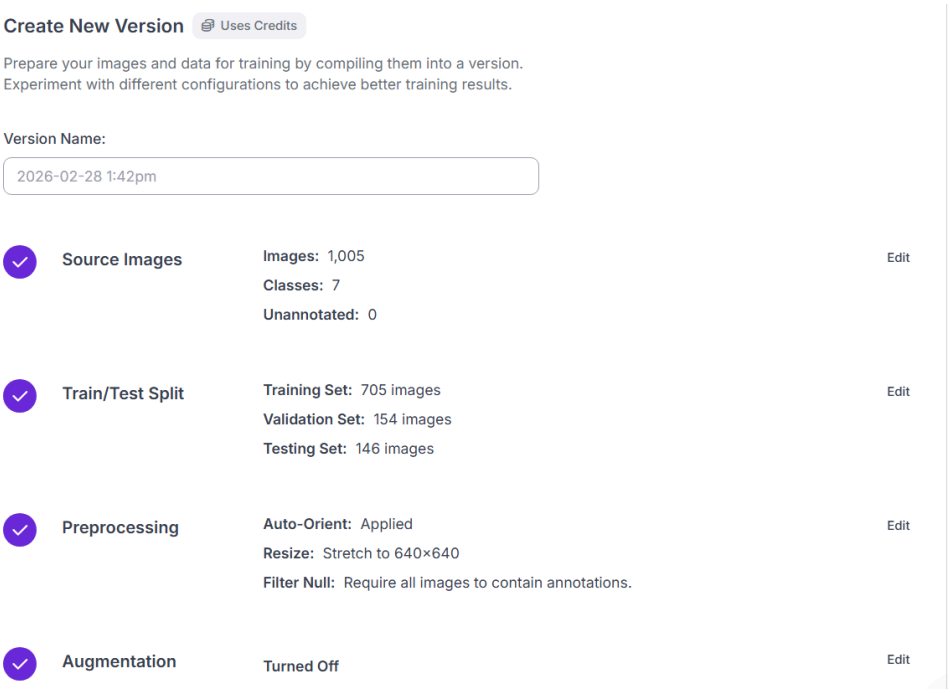
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Dataset Version and Export



The initial annotated release comprises 1,005 labeled frames distributed across the seven defined classes, with no unannotated images included in the export. To enforce dataset integrity and prevent the introduction of unlabeled frames into the training pipeline—which could produce silent class underrepresentation or missing-label errors during model training—a Filter Null condition was applied during version generation. This configuration requires that all images included in the version carry at least one annotation, ensuring a fully labeled export.

The dataset was partitioned into a standard three-way split to support model development and performance evaluation. Of the 1,005 frames, 705 were allocated to the training subset, 154 to the validation subset, and 146 to the test subset. The training split is used for parameter optimization during model fitting; the validation split supports iterative model selection and hyperparameter tuning; and the test split is held out for final, unbiased performance reporting. This partition structure conforms to standard practice for producing reliable estimates of detector generalization (Redmon et al., 2016; Everingham et al., 2010).

Preprocessing was applied uniformly across all images to standardize input dimensions and correct orientation artifacts. The auto-orient function was enabled to resolve orientation inconsistencies that may arise from differences in capture metadata or encoding. All frames were subsequently resized to 640×640 pixels using a stretching transformation, establishing a uniform input resolution compatible with standard YOLO

architectures. In this release, no augmentation was applied, meaning the exported dataset consists exclusively of the original extracted and preprocessed frames. This conservative configuration was chosen to maintain interpretability: by excluding synthetic transformations, model behavior can be more directly attributed to the properties of the real-world captured data rather than to augmentation artifacts.

The final dataset version was exported in a YOLO-compatible object detection format, in which each image is paired with a corresponding plain-text label file containing normalized bounding box coordinates and integer class identifiers. This export structure enables direct integration into standard YOLO training workflows and preserves a complete record of the split configuration, preprocessing steps, and version metadata associated with this release.

Below is a sample image of the final dataset version exported:



V. References

Bochkovskiy, A., Zhaksybekov, A., & Abdullaev, O. (2024). Domain generalization in autonomous driving: Evaluating YOLOv8s, RT-DETR, and YOLO-NAS with the ROAD-Almaty dataset. arXiv. https://arxiv.org/abs/2412.12349

Everingham, M., Van Gool, L., Williams, C. K. I., Winn, J., & Zisserman, A. (2010). The Pascal Visual Object Classes (VOC) challenge. International Journal of Computer Vision, 88(2), 303–338. https://doi.org/10.1007/s11263-009-0275-4

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 779–788. https://arxiv.org/abs/1506.02640

Roboflow. (2024). Dataset versions. Roboflow Documentation. https://docs.roboflow.com/datasets/dataset-versions

Vasquez Hidalgo, C., Rahnama, A., Orekondy, T., & Bhatt, U. (2024). Classification drives geographic bias in street scene segmentation. arXiv. https://arxiv.org/abs/2412.11061